

Yuan Jiang

Ph.D. Candidate, Industrial & Enterprise Systems Engineering
University of Illinois Urbana-Champaign, IL, USA

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My research aims to develop physics-informed digital twins for safety-critical engineering systems, integrating multi-physics modeling, data-efficient machine learning, and uncertainty-aware decision-making. I focus on energy storage, aerospace propulsion, additive manufacturing, and electro-mechanical systems, with applications in reliability assessment, prognostics, condition monitoring, and design optimization.

Education

University of Illinois Urbana-Champaign

Ph.D., Industrial Engineering

Advisor: Prof. Pingfeng Wang

Dissertation: Physics-informed multi-fidelity digital twins for design, diagnosis, and prognosis of complex engineering systems

Urbana, IL, USA

Aug 2023 – May 2027 (Expected)

Tongji University

M.Sc., Vehicle Engineering

Advisor: Prof. Gang Niu

Thesis: Time-frequency feature extraction and health monitoring of rail short-pitch damage

Shanghai, China

Sep 2020 – Jun 2023

Tongji University

B.Sc., Vehicle Engineering

Shanghai, China

Sep 2015 – Jul 2020

Work Experience

NIO Automobile, Inc.

Power System Intern

Developed data-driven condition monitoring and fault detection methods for EV power swap and charging systems, translating large-scale operational and sensor data into real-time health indicators and early-warning signals.

Shanghai, China

May – Aug 2023

HiRain Technologies

Autonomous Driving Development Intern

Contributed to perception software development and data processing pipelines for multi-camera autonomous driving systems, with exposure to large-scale, high-throughput data integration and real-time system constraints.

Shanghai, China

Jun – Aug 2022

NIO Automobile, Inc.

Battery Testing Intern

Conducted experimental testing and data analysis for lithium-ion batteries, supporting model validation and data-informed performance assessment under real-world operating conditions.

Shanghai, China

Feb – May 2021

Research Experience

Physics-guided Multi-fidelity Digital Twins for Wet Clutch Diagnostics

Sponsor: REMADE Institute / John Deere Collaboration

- Led the development of multi-fidelity fluid–structure interaction (FSI) digital twins for wet clutch systems by integrating high-fidelity contact-pair simulations with computationally efficient spring–damper approximations
- Proposed a physics-guided Fourier deep compositional neural operator with two-stage multi-fidelity training strategy for transient hydraulic pressure prediction under clutch wear and missing-plate fault conditions

Physics-informed Digital Twin Modeling for Hall Thruster Degradation and Prognostics

Sponsor: Air Force Office of Scientific Research (AFOSR)

- Led the development of a multi-fidelity digital twin framework that integrates physics-based plasma simulations and physics-informed neural surrogates for data-efficient modeling of Hall thruster erosion and degradation under sparse observations
- Proposed uncertainty-aware prognostic methodologies that couple physics-informed surrogate models with Bayesian state estimation, enabling online update and remaining useful life prediction under variable operating and environmental conditions

Physics-informed Machine Learning for Battery Thermal-Health Management and Design Optimization

Sponsor: National Science Foundation (NSF)

- Led the development of physics-informed neural operator and CNN-based surrogate models for data-efficient, multi-fidelity thermal field modeling in lithium-ion battery packs with complex geometries
- Formulated a unified learning–optimization framework that couples learned thermal surrogates with reliability-aware design optimization, enabling robust layout-level decision-making under data and model uncertainty

- Provided technical mentoring and methodological guidance to junior researchers and collaborators, supporting the application of physics-informed modeling frameworks to battery thermal runaway and transmission system diagnostics

Data-driven Signal Processing for Non-stationary System Diagnostics

Sponsor: National Science Foundation (NSF)

- Led the development of data-driven time-frequency signal processing methodologies for diagnostics of non-stationary systems under variable operating conditions, establishing robust and physically interpretable feature extraction from noisy sensor data
- Established a methodological bridge between classical signal processing and data-driven modeling by formalizing signal representations that interface naturally with prognostic and health monitoring workflows

Publications

indicates equal contribution

11 published peer-reviewed journal papers (6 first-author), including IEEE TII, RESS, MSSP, EAAI, and ASME JMD.

Journal Papers

- [J1] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, and Pingfeng Wang. Physics-informed cross-attention operator network with hard-constrained Fourier features for heat map prediction of large-scale battery packs. *Engineering Applications of Artificial Intelligence*, 176 (2026) 114833. DOI: [10.1016/j.engappai.2026.114833](https://doi.org/10.1016/j.engappai.2026.114833).
- Second Place (Poster Competition), 2026 ISE Student Conference (2nd out of 28)
- [J2] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, Yuejian Chen, and Pingfeng Wang. Iterative dispersive Vold-Kalman filter with local adaptive bandwidth for dispersive signal decomposition in structural health monitoring. *IEEE Transactions on Industrial Informatics*. (Accepted)
- [J3] Yiyue Jiang, **Yuan Jiang**, and Pingfeng Wang. A review on mitigating thermal runaway propagation in battery packs: From mechanisms to modeling and design optimization. *Energy Advances*, (2026). DOI: [10.1039/D5YA00351B](https://doi.org/10.1039/D5YA00351B)
- [J4] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang. Remaining useful life prediction for Hall thrusters based on adaptive self-cognizant dynamic system and multi-physics modeling. *Journal of Mechanical Design*, 148 (7) (2026) 071701. DOI: [10.1115/1.4070967](https://doi.org/10.1115/1.4070967).
- ASME DAC Paper of Distinction (Top 10 out of 103)
- [J5] Zheng Liu, Yanwen Xu, **Yuan Jiang**, Anabel Renteria, Parth Bansal, Chenlong Xu, Pingfeng Wang, and Yumeng Li. Uncertainty quantification of additively manufactured architected cellular materials for energy absorption applications. *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering*, 11 (3) (2025). DOI: [10.1115/1.4066933](https://doi.org/10.1115/1.4066933).
- [J6] Pouya Kabirzadeh, Zheng Liu, Mostafa Olyaei, Haoyun Qiu, Yashraj Gurumukhi, Harsh Tyagi, **Yuan Jiang**, Vivek S Garimella, Bakhshish Preet Singh, Yumeng Li, Pingfeng Wang, and Nenad Miljkovic. Integrating heat transfer and control optimization: A comprehensive review of battery thermal management systems. *Journal of Energy Storage*, 131 (2025) 117289. DOI: [10.1016/j.est.2025.117289](https://doi.org/10.1016/j.est.2025.117289).
- [J7] **Yuan Jiang**[#], Zheng Liu[#], Pouya Kabirzadeh, Yulun Wu, Yumeng Li, Nenad Miljkovic, and Pingfeng Wang. Multi-fidelity physics-informed convolutional neural network for heat map prediction of battery packs. *Reliability Engineering & System Safety*, 256 (2025) 110752. DOI: [10.1016/j.ress.2024.110752](https://doi.org/10.1016/j.ress.2024.110752).
- [J8] Yuejian Chen, Zihan Li, **Yuan Jiang**, Chunsheng Yang, Min Xia, and Ke Feng. Enhanced sparse LPV-ARMA model with ensemble basis functions for mechatronic transmission fault detection under variable speed conditions. *IEEE Internet of Things Journal*, 12 (11) (2025) 17223-17232. DOI: [10.1109/JIOT.2025.3535580](https://doi.org/10.1109/JIOT.2025.3535580).
- [J9] Yuejian Chen, Zihan Li, **Yuan Jiang**, Dao Gong, and Kai Zhou. Sparse LPV-ARMA model for non-stationary vibration representation and its application on gearbox tooth crack detection under variable speed conditions. *Mechanical Systems and Signal Processing*, 224 (2025) 112161. DOI: [10.1016/j.ymssp.2024.112161](https://doi.org/10.1016/j.ymssp.2024.112161).
- [J10] **Yuan Jiang**, Yuejian Chen, and Pingfeng Wang. An iterative adaptive Vold-Kalman filter for non-stationary signal decomposition in mechatronic transmission fault diagnosis under variable speed conditions. *IEEE Transactions on Industrial Informatics*, 20 (8) (2024) 10510-10519. DOI: [10.1109/TII.2024.3393536](https://doi.org/10.1109/TII.2024.3393536).
- [J11] **Yuan Jiang** and Gang Niu. An iterative frequency-domain envelope-tracking filter for dispersive signal decomposition in structural health monitoring. *Mechanical Systems and Signal Processing*, 179 (2022) 109329. DOI: [10.1016/j.ymssp.2022.109329](https://doi.org/10.1016/j.ymssp.2022.109329).

Journal Papers under Review

- [J12] **Yuan Jiang**[#], Parth Bansal[#], Mohammad Mundiwalla, Chao Hu, and Pingfeng Wang. Physics-guided multi-fidelity neural operator digital twin for wet clutch hydraulic pressure prediction. *Journal of Mechanical Design*. (Under first round of review)

Conference Proceedings

- [C1] **Yuan Jiang**, and Pingfeng Wang. Physics-informed self-cross attention operator network for heat map prediction in large-scale battery thermal management systems. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)

- [C2] Jiayi Zhao[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang. A digital-twin-based framework for inverse diagnosis of wet clutch hydraulic pressure signals using time-series similarity matching. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)
- [C3] Yulun Wu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang. From local to global: CNN surrogates for scaling layout-resolved temperature prediction in battery thermal management. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)
- [C4] **Yuan Jiang**, and Pingfeng Wang. Multi-fidelity adaptive self-cognizant dynamic system for remaining useful life prediction of Hall thrusters. *2026 IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)
 - IISE QCRE Best Student Poster Award (1st out of 18)
- [C5] Yiyue Jiang[#], **Yuan Jiang**[#], and Pingfeng Wang. Spatiotemporal design-aware surrogate modeling of thermal runaway propagation in Li-ion battery packs. *2026 IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)
 - IISE M&D Best Track Paper Finalist (Top 4 out of 40)
- [C6] Yanbin Luo, **Yuan Jiang**, and Pingfeng Wang. Degradation behavior and health monitoring of GaN HEMTs under high-temperature stress conditions. *2026 IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)
- [C7] Jiayi Zhao, **Yuan Jiang**, Yiyue Jiang, and Pingfeng Wang. A complexity-aligned benchmark for adaptive sampling in GP global fitting. *2026 IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)
- [C8] Alexandra N. Leeming, **Yuan Jiang**, Stephen Messing, Siyuan Wang, Pingfeng Wang, J. Gary Eden, Peter D. Dragic, Joshua L. Rovey, Valentin Korman, and Kurt Polzin, Exploration of an in-situ Hall thruster erosion and plasma diagnostic sensor, *2026 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum)*, Orlando, FL, USA. 12 Jan. 2026, 14 Pages. DOI: [10.2514/6.2026-0929](https://doi.org/10.2514/6.2026-0929)
- [C9] Zheng Liu[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang, Physics-informed machine learning enhanced battery pack optimization, *2025 IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+ EATS)*, Anaheim, CA, USA. 18 June 2025, 5 Pages. DOI: [10.1109/ITEC63604.2025.11098093](https://doi.org/10.1109/ITEC63604.2025.11098093).
- [C10] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang, Prognostics of Hall thruster erosion using multiphysics-based modeling and machine learning, *2025 Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: [10.1109/RAMS48127.2025.10935282](https://doi.org/10.1109/RAMS48127.2025.10935282).
- [C11] Zheng Liu[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang, Physics-informed machine learning for battery pack thermal management, *2025 Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: [10.1109/RAMS48127.2025.10935157](https://doi.org/10.1109/RAMS48127.2025.10935157).
- [C12] Shimeng Yang, **Yuan Jiang**, Zheng Liu, and Pingfeng Wang, Transductive transfer learning features for prognostics and health management, *2025 Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: [10.1109/RAMS48127.2025.10935143](https://doi.org/10.1109/RAMS48127.2025.10935143).
- [C13] Parth Bansal, **Yuan Jiang**, Zhou Li, Sergio Cordero, Zahra Heussen, Debbie Senesky, Pingfeng Wang, and Yumeng Li, Multiphysics modeling and simulation of gas sensor for NO₂ detection, *2024 ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Portland, OR, USA, 17 Nov. 2024, 6 Pages. DOI: [10.1115/IMECE2024-145663](https://doi.org/10.1115/IMECE2024-145663).
- [C14] **Yuan Jiang** and Gang Niu, Rail local damage detection based on recursive frequency-domain envelope tracking filter and rail impact index, *2022 Global Reliability and Prognostics and Health Management (PHM 2022)*, Yantai, China. 13 Oct. 2022, 7 Pages. DOI: [10.1109/PHM-Yantai55411.2022.9942082](https://doi.org/10.1109/PHM-Yantai55411.2022.9942082).
 - IEEE PHM Best Paper Award (Top 5 out of 274)
- [C15] Hongyang Zhao, **Yuan Jiang**, and Gang Niu, A tacholeless order tracking method based on extended intrinsic chirp component decomposition for gears under large speed variation conditions, *Journal of Physics: Conference Series*, 2184 (1) (2022) 12052. DOI: [10.1088/1742-6596/2184/1/012052](https://doi.org/10.1088/1742-6596/2184/1/012052).
- [C16] **Yuan Jiang**, Hongyang Zhao, and Gang Niu, Intelligent rolling bearing fault diagnosis under variable speed conditions without tachometers, *2021 Global Reliability and Prognostics and Health Management (PHM 2021)*, Nanjing, China. 15 Oct. 2021, 7 Pages. DOI: [10.1109/PHM-Nanjing52125.2021.9613049](https://doi.org/10.1109/PHM-Nanjing52125.2021.9613049).

Teaching & Mentoring Experience

Teaching interests include reliability engineering, signal processing, engineering design analysis, design optimization, and decision-making under uncertainty. At the graduate level, I am interested in physics-informed machine learning, digital twin modeling, and data-driven design optimization for safety-critical engineering systems.

Teaching Assistant

SE 411: Reliability Engineering

Main instructor: Prof. Krishnan Girish

University of Illinois Urbana-Champaign

Spring 2025

- Delivered guest lectures on reliability analysis and reliability-based design optimization, covering problem formulation, algorithmic implementation, and engineering interpretation.
- Designed an RBDO course project and mentored students on modeling, optimization algorithms, and coding implementation.
- Led exam review sessions with structured problem-solving tutorials to reinforce core concepts and analytical skills.

SE 450: Decision Analysis

Main instructor: Prof. Pingfeng Wang

University of Illinois Urbana-Champaign

Fall 2024, Fall 2025

- Contributed to undergraduate instruction by leading discussion sections and providing academic support via office hours.
- Delivered exam review lectures covering multivariate decision making, decision trees, and uncertainty-based decision making, with an emphasis on systematic problem formulation and solution strategies.

Rail Vehicle Systems Health Management

Main instructor: Prof. Gang Niu (Instructed in English)

Tongji University

Spring 2022

- Supported course delivery by assisting lectures and coordinating laboratory components for an undergraduate course.
- Led laboratory sessions on bearing and gearbox test rigs, covering experimental setup, data acquisition, and signal processing.
- Instructed students on vibration-based fault diagnosis using FFT, wavelet transforms, and machine learning methods.

Student Mentoring

Yanbin Luo (M.S., UIUC)

Jul 2025 – Present

Experimental Degradation Modeling of High-Electron-Mobility Transistors (HEMTs) under Extreme Operating Conditions

- Primary technical advisor for experimental design, data processing, and degradation modeling.
- Guided master's thesis and IISE conference paper preparation (accepted).

Jingsen Yang (M.S., UIUC)

Nov 2025 – Present

Multi-fidelity Surrogate Modeling and Data Fusion based on Foundational Tabular Model

- Primary technical advisor for surrogate modeling, data processing, and methodological development.
- Thesis and conference paper in preparation.

Awards & Honors

Research Awards

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| 1. IISE QCRE Best Student Poster Award, IISE Annual Conference & Expo | 2026 |
| 2. IISE M&D Best Track Paper Finalist, IISE Annual Conference & Expo | 2026 |
| 3. Second Place (Poster Competition), ISE Student Conference, University of Illinois Urbana-Champaign | 2026 |
| 4. ASME-DAC Paper of Distinction, Design Automation Committee (DAC), ASME | 2025 |
| 5. Outstanding Master's Thesis Award, Tongji University | 2023 |
| 6. IEEE PHM Best Paper Award, IEEE Reliability Society | 2022 |

Fellowships & Scholarships

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| 1. Mavis Future Faculty Fellow, University of Illinois Urbana-Champaign | 2026 |
| 2. Jerry S. Dobrovolsky Fellowship, University of Illinois Urbana-Champaign | 2023 |
| 3. National Scholarship of China, Ministry of Education of the People's Republic of China | 2019, 2022 |

Professional Recognition

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| 1. IOE-ISyE-MS&E Joint Rising Star, IOE (UMich), ISyE (Georgia Tech), MS&E (Stanford) | 2026 |
| 2. IEEE TIM Outstanding Reviewer, IEEE Instrumentation and Measurement Society | 2025 |

Academic Service

Conference Leadership

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| • Session Chair, Prognostics and Reliability in Power and Energy Systems, IISE Annual Conference & Expo | 2026 |
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Journal Reviewer

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|---|------------------------------------|
| • Reliability Engineering & System Safety | • Neurocomputing |
| • Engineering Applications of Artificial Intelligence | • Journal of Process Control |
| • IEEE Transactions on Reliability | • Journal of Vibration and Control |

- Mechanical Systems and Signal Processing
- IEEE Transactions on Instrumentation & Measurement
- ISA Transactions
- IEEE Sensors Journal
- Computers & Structures
- Journal of Rail and Rapid Transit

Conference Reviewer

- ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE)

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- American Society of Mechanical Engineers (ASME)
- Institute of Industrial and Systems Engineers (IISE)
- Institute for Operations Research and the Management Sciences (INFORMS)